

Where the Loops Come From

Why some networks branch and others join back up

Reading and note-making task • Year 9 • Chemistry, Biology & Geography

Part A Before you read: draw and predict

Do not read the essay yet.

A1. In the boxes below, sketch each network from memory. Do not worry about accuracy; worry about whether lines ever join back together.

A river and its streams	The veins in a leaf	The streets near school

A2. In which of your three sketches does a line ever leave a point and come back to it? Circle them.

A3. Building an extra connection always costs extra material. So why would anything ever build one? Your best guess:

Keep this page. You will mark your own prediction at the end.

Glossary strip use while reading

network	any set of points joined by links: streams, streets, tubes, roots
tree (in maths)	a network where no route ever returns to where it started
loop	a route that does return, making a ring in the network
slime mould	a single huge cell that spreads out and finds food without a brain
fungal network	threads growing through soil, connecting to tree roots
redundant	spare; not needed unless something else fails
predictable	known in advance, the same tomorrow as today

Part B Cornell notes: five networks and one rule

Fill the **Notes** column as you read. Some cues are given; write your own in the blank cue lines.

Cue 1. FROST Why does a frost arm that pokes ahead grow even faster? What makes frost different from every other network here?	Notes
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Cue 2. THE COST OF A LOOP Why is a loop always more expensive than no loop? <i>Your own cue:</i> _____	Notes
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Cue 3. RIVERS Three things about a river's job that never change. List them. So what shape does it end up as?	Notes
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Cue 4. SLIME MOULD Two steps in its method: flood, then ...? What three things were measured in the Tokyo comparison?	Notes
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Cue 5. THE RULE Complete both halves: predictable traffic → changing traffic → <i>Your own cue:</i> _____	Notes
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Cue 6. FUNGI, AND THE WARNING What is definitely true about fungal networks, and what is doubtful? <i>Your own cue:</i> _____	Notes
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Cue 7. CITIES What is a city missing that the slime mould has? <i>Your own cue:</i> _____	Notes
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Summary box. In no more than **40 words**, and without using the words *river*, *slime* or *city*, state the rule this essay is about.

Part C Cloze: the argument in one paragraph

Fill each gap. Every answer is one or two words.

Frost branches, but nothing ever _____ along it, so its shape answers only to the water around it. Every other network here has a job, and that job costs material.

A network with no rings in it is called a _____. Adding a ring means adding a _____, which always costs more to build.

A river never builds one, because its traffic only goes one way, towards one destination, and tomorrow is the same as today. When the job is that _____, the shortest network wins.

A slime mould does build a few. It first _____ everywhere, then thickens the tubes carrying the most and lets the rest wither. It keeps spare links because it cannot know where the next _____ will appear.

So the rule is this: loops are worth their cost only when the traffic keeps _____, or when a link might get _____. Fungi and cities both face exactly that, which is why both of them are full of rings.

Part D Understanding and reflection

1. **Sorting task.** Predict whether each of these has loops or not, then say what its traffic is. *Lightning, the London Underground, the veins in a leaf, a tree's branches, blood capillaries, a drainage ditch system.*

Tree: no loops	Has loops

Pick the one you were least sure about and explain what made it difficult.

2. **Design a test.** A friend claims that every network built by a living thing has loops in it. Using only things in this essay, give one example that proves them wrong, and explain why that example does not break the rule.

3. **The careful bit.** The essay says the popular story about the wood-wide web is stronger than the evidence supports. Write down what is agreed and what is disputed, in your own words, in two lines.

4. **Push the idea further.** Think of a network built by people that changed from a tree into a loop, or the other way round, when the traffic on it changed. Describe it.

5. **Back to your prediction.** Reread your answer to A3. What did you get right, and what would you now change?

Teacher key and notes

Part C cloze. travels; tree; link (accept “connection”); predictable; floods; food (accept “oat flake”); changing; broken (accept “cut”).

Part D1 sorting. No loops: lightning, a tree’s branches, a drainage ditch system. Loops: the London Underground, the veins in a leaf, blood capillaries. The two productive difficulties are the leaf, which most students draw as a tree and which is in fact heavily looped, and lightning, which branches without carrying repeated traffic at all and so belongs with frost rather than with rivers. A tree’s branches are the cleanest tree on the list, since sap moves one way to a known set of destinations.

Part D2. The intended answer is the river, or a tree’s branches, or lightning. The rule survives because none of them faces unpredictable or reversing traffic; the claim to defeat is “living thing” rather than the rule itself.

Part D4. Strong answers include a railway that gained a loop line as commuting patterns spread, an electricity grid moving from radial to meshed supply, or a village lane network absorbed into a ring road. Accept anything where the change in traffic is stated, not just the change in shape.

Pitching note. This edition matches the Year 9 version of the essay. Cue boxes 1–7 follow its sections in order. Section 5, the rule, is the hinge; if time is short, cut the fungal section rather than that one, since cities alone are enough to complete the argument. The fungal section is the only place where students meet a contested claim, and it repays being read aloud together.

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